

Interfacing CouchDB with Python

Apache CouchDB is a database that is focused on ease of use. It can be interfaced with the versatile Python programming language, as shown in this tutorial.

Apache CouchDB is a popular open source database that is used as a document-oriented NoSQL database.

Like most other NoSQL databases, CouchDB uses the JSON format to store data. JSON (JavaScript Object Notation) is the open standard data file format that is used as an alternate to XML, to transmit data between multiple incompatible and heterogeneous servers.

Key features of CouchDB

- It has a distributed architecture with easy and fast replication of a database for multiple instances of the server
- Quick retrieval of data and indexing
- Interface based on REST for insertion, updates, retrieval and deletion of documents
- JSON-based document format (easily translatable across different languages)
- Assorted libraries available for multiple languages, or simply for the localisation and internationalisation of content
- Subscription of data updates is very rapid
- Document storage
- ACID based semantics
- MapReduce indexes and views
- All items have a unique URI for HTTP
- Consistency

- Replication for devices, including smartphones for handling offline and online data sync
- In-built administration panel that can be accessed using a Web GUI admin panel called Futon
- Lock-free database updates
- Free-form document format

Installing CouchDB

To install on Ubuntu/Debian Linux, type:

```
$ sudo aptitude install couchdb
```

To install on Gentoo Linux, type:

```
$ sudo emerge couchdb
```

Services can be started or stopped using the *init* scripts in all the distributions as follows:

```
$ /etc/init.d/couchdb start
```

For Windows, the CouchDB installer is available on <http://couchdb.apache.org>. The CouchDB installed on the system can be executed in standalone mode as well as in service mode.



CouchDB

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